

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

PRELAB EXERCISES (2 points)*This prelab must be completed and handed in to the lab TA at the start of the lab.***Question 1****1 point**

You will be using a diffraction grating in this lab exercise as a dispersive element in a spectrometer. When you begin to examine the Balmer series of atomic hydrogen, you will observe an indigo line, a red line, and a violet line as you move the spectrometer's telescope away from the zero angle (0th order) position.

What will be the sequence of the spectral lines, starting from the zero angle position? Explain why, showing some calculations or a diagram.

Question 2**1 point**

Showing all work, what is the expected measured grating separation, d , if you use:

(a) ($\frac{1}{2}$ point) A 600 groove/mm grating?

(b) ($\frac{1}{2}$ point) A 300 groove/mm grating?